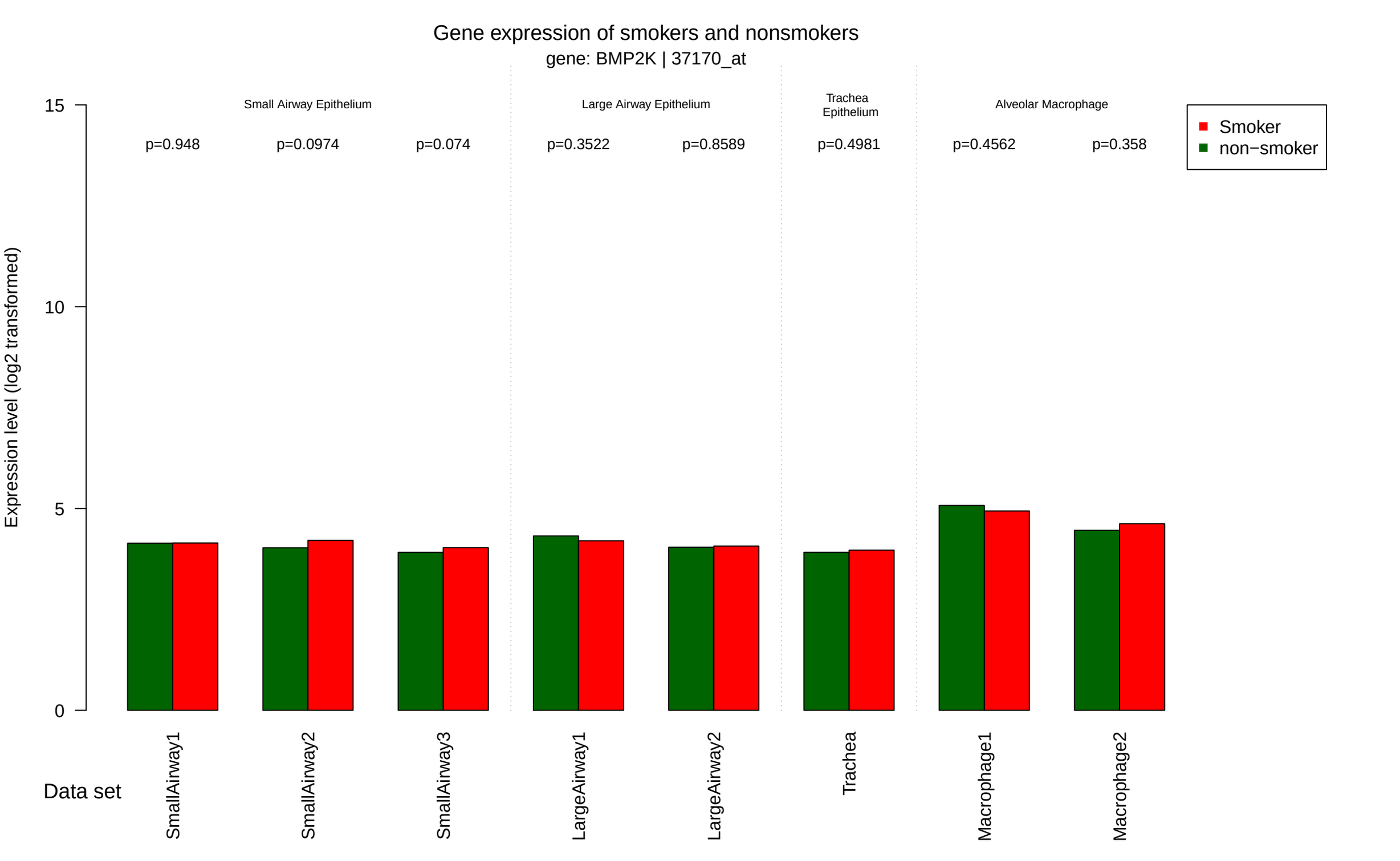
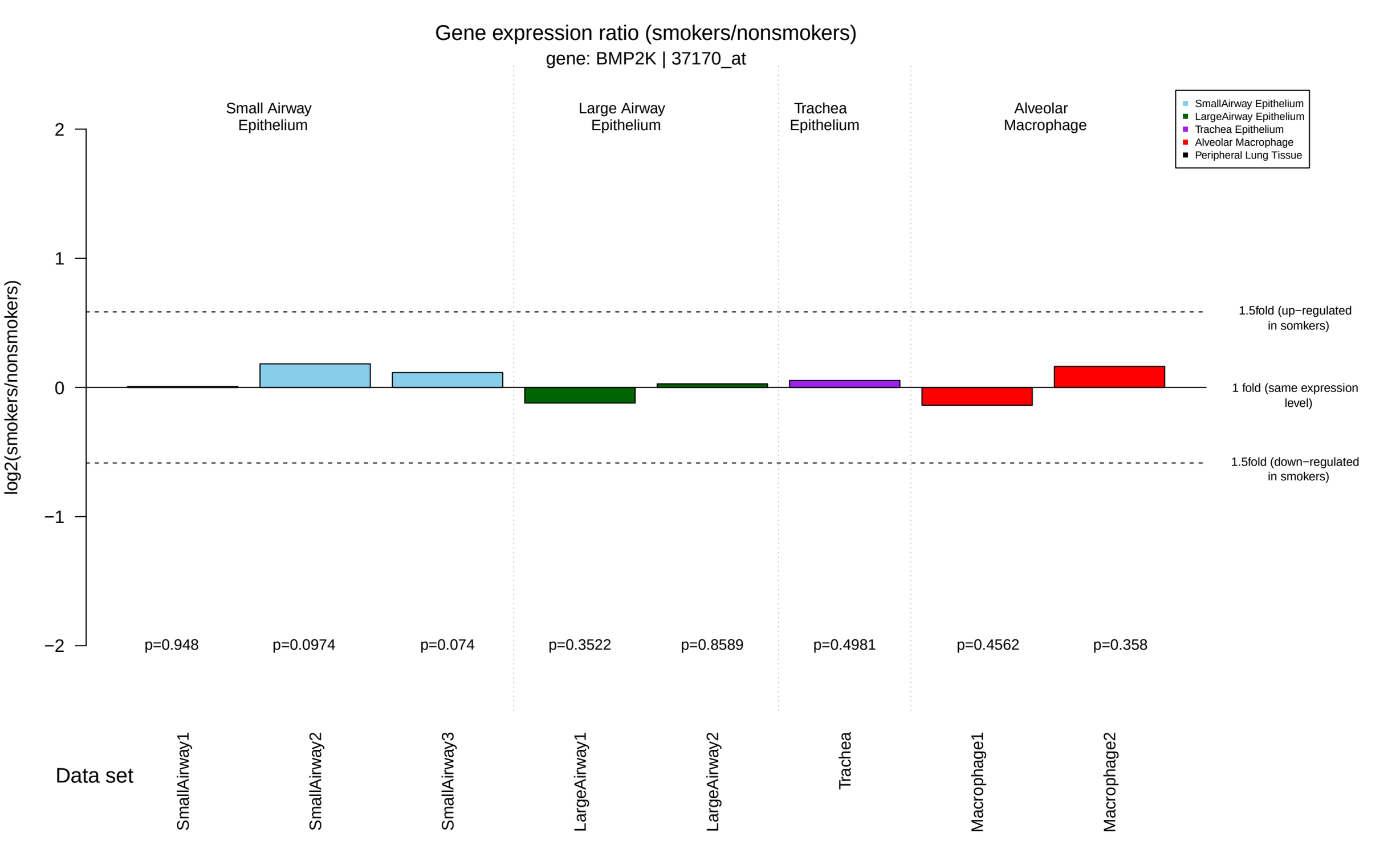
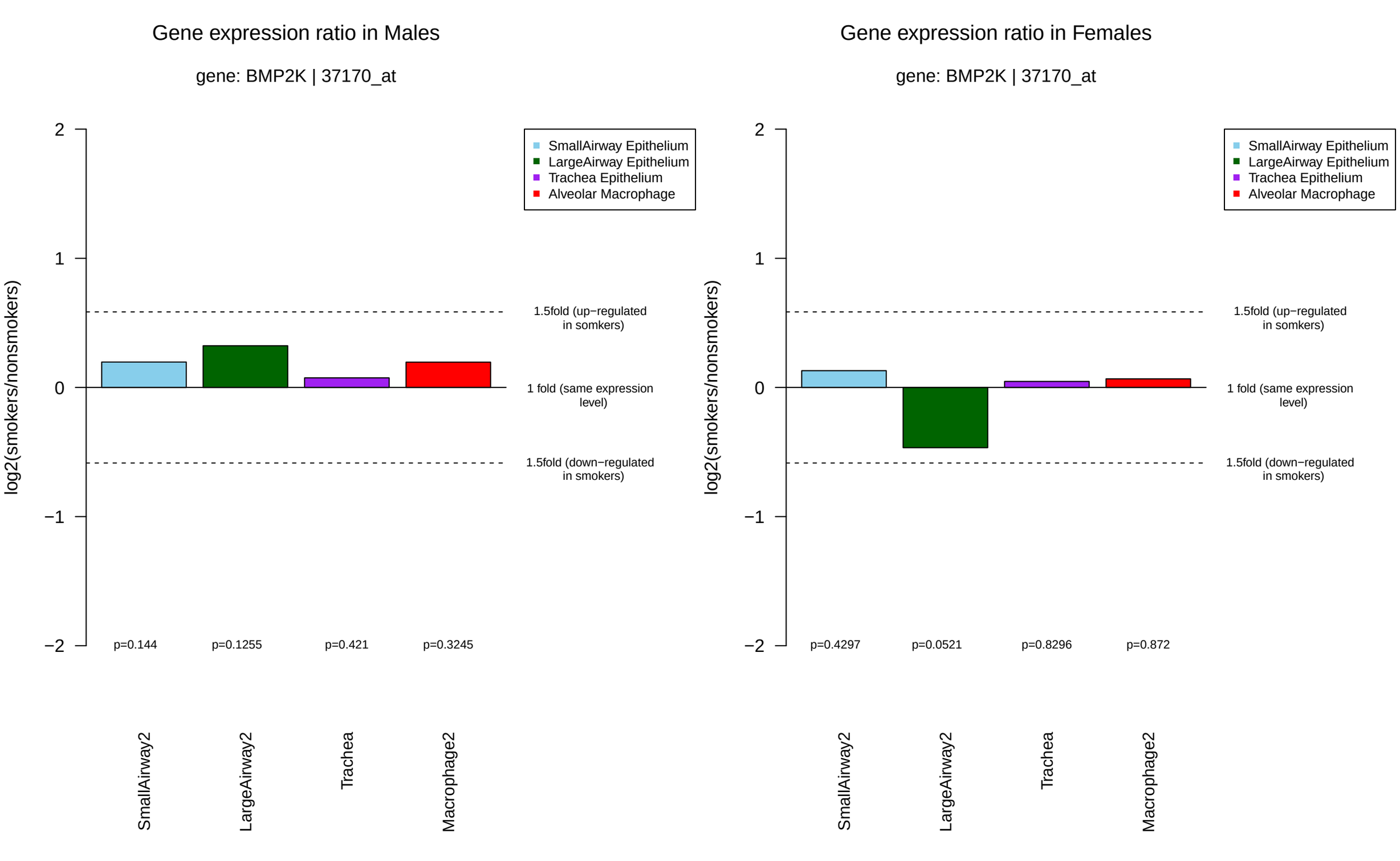
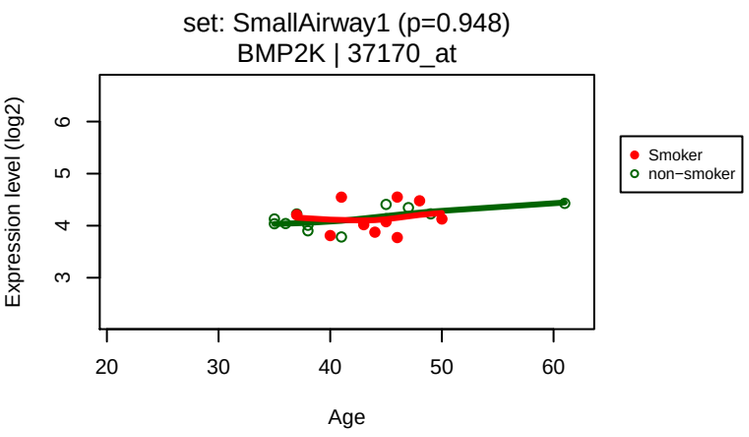
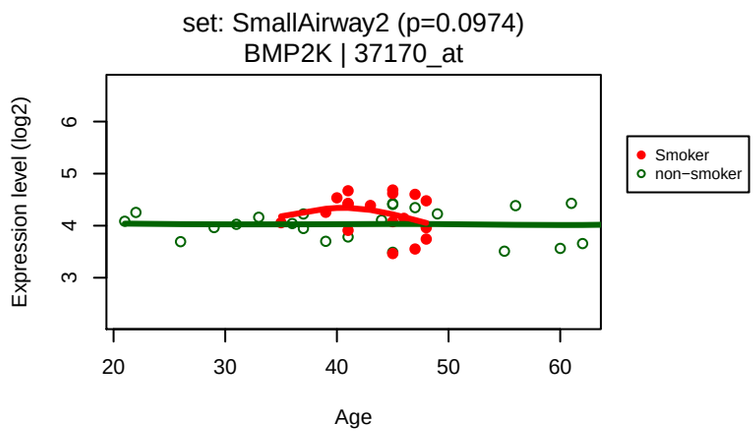
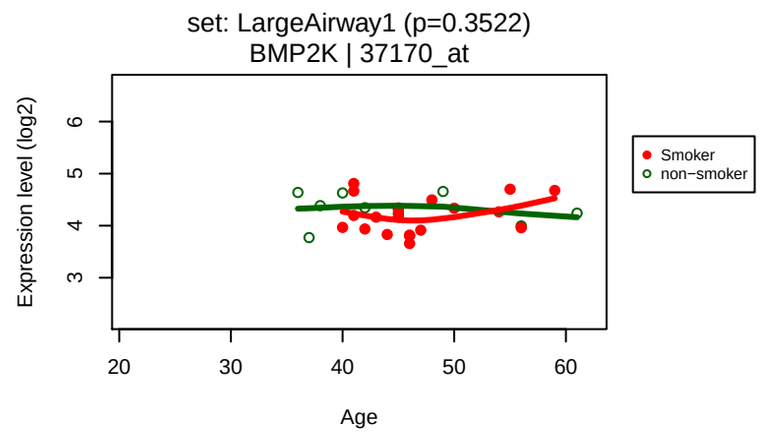

Gene expression of smokers and nonsmokers
gene: BMP2K | 37170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.948
p=0.0974
p=0.074
p=0.3522
p=0.8589
p=0.4981
p=0.4562
p=0.358
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

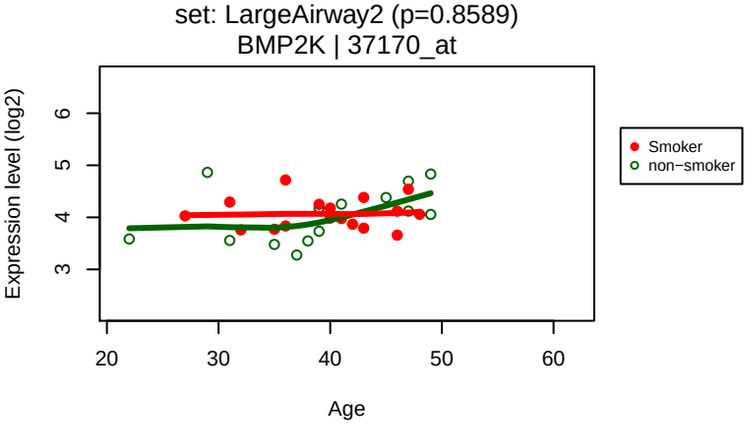

Gene expression ratio (smokers/nonsmokers)
gene: BMP2K | 37170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.948
p=0.0974
p=0.074
p=0.3522
p=0.8589
p=0.4981
p=0.4562
p=0.358
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

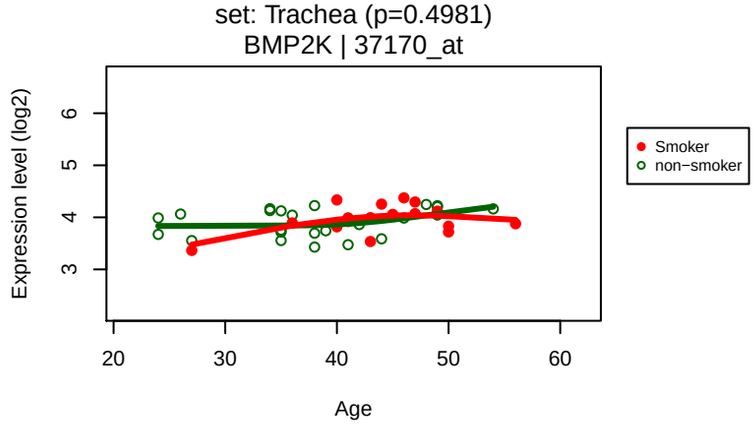

Gene expression ratio in Males
gene: BMP2K | 37170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.144
p=0.1255
p=0.421
p=0.3245
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP2K | 37170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4297
p=0.0521
p=0.8296
p=0.872
SmallAirway2
LargeAirway2
Trachea
Macrophage2

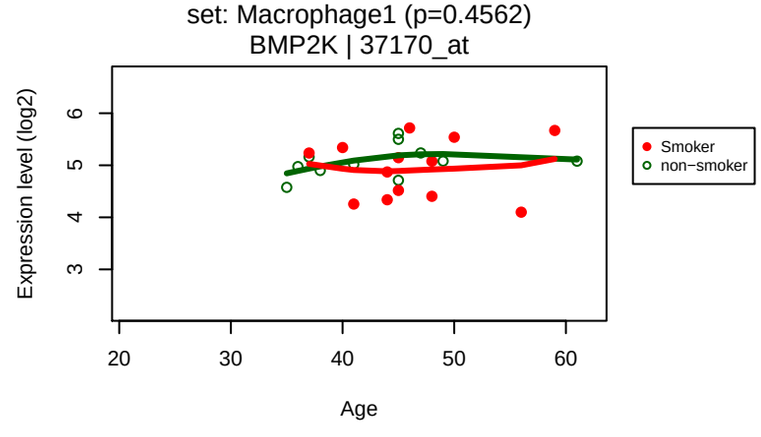

set: SmallAirway1 (p=0.948)
BMP2K | 37170_at
Expression level (log2)
Age
Smoker
non−smoker

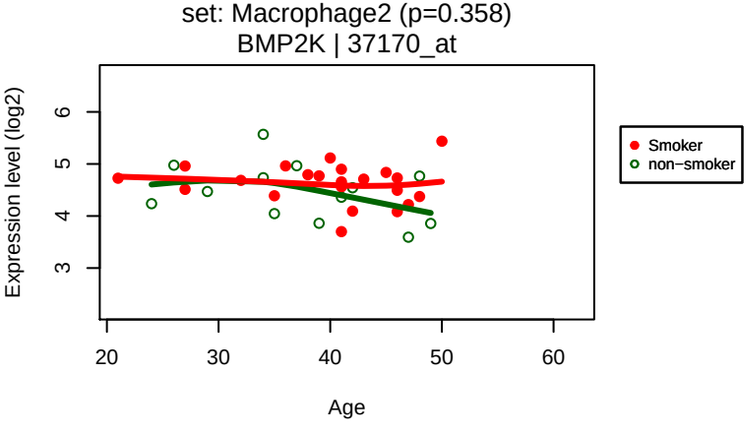

set: SmallAirway2 (p=0.0974)
BMP2K | 37170_at
Expression level (log2)
Age
Smoker
non−smoker

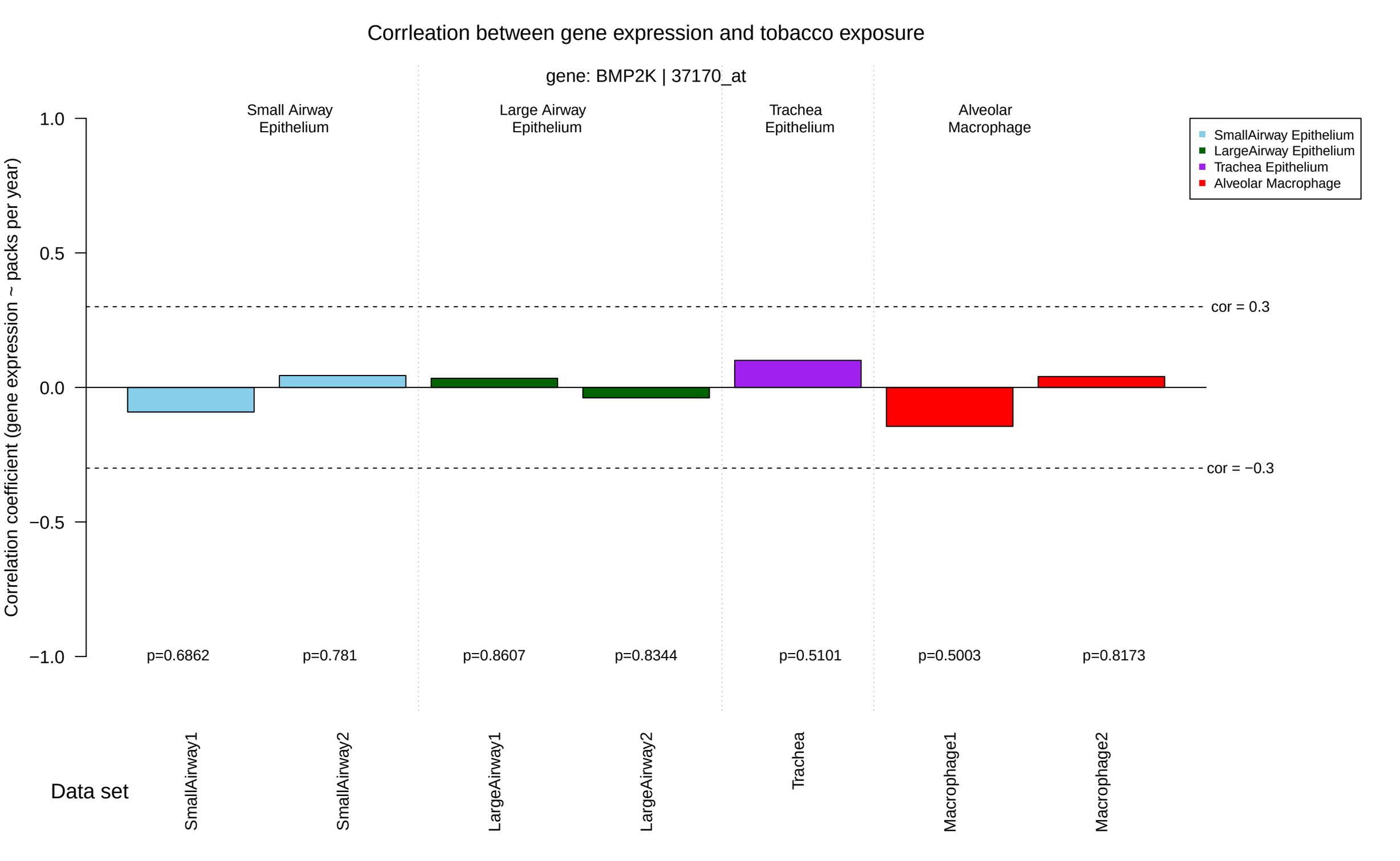

set: LargeAirway1 (p=0.3522)
BMP2K | 37170_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8589)
BMP2K | 37170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4981)
BMP2K | 37170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4562)
BMP2K | 37170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.358)
BMP2K | 37170_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP2K | 37170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6862
p=0.781
p=0.8607
p=0.8344
p=0.5101
p=0.5003
p=0.8173
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2